



SIMATS ENGINEERING

SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES



Dept. Name: CSE - AIDS

Course Code/Course Name: DSA06 /Data Handling and Visualization

LIST OF EXPERIMENTS

Set 1: Monthly Sales Data

Dataset: Monthly Sales

1. Create a line chart to visualize the monthly sales. Label the axes and title the chart appropriately.
2. Generate a bar chart to display the top-selling products for the year. Label the chart elements.
3. Develop a scatter plot to explore the relationship between advertising budget and monthly sales. Explain the insights drawn from the scatter plot.
4. Build an interactive dashboard combining the line chart and bar chart to allow users to explore sales data interactively.

Set

2:

Month	Sales (in \$)
January	15000
February	18000
March	22000
April	20000
May	23000

Customer Feedback Analysis

Dataset: Customer Satisfaction

Customer ID	Age	Satisfaction Score
1	25	4
2	30	5
3	35	3
4	28	4
5	40	5

1. Create a histogram to represent the distribution of customer ages. Label the axes and title the chart.
2. Generate a pie chart to display the overall distribution of customer satisfaction scores. Include labels.
3. Build a stacked bar chart to visualize the distribution of customer satisfaction scores by age group.
4. Develop a word cloud from open-ended customer feedback to identify prevalent customer sentiments.

Set 3: Employee Performance Evaluation

Dataset: Employee Performance

Employee ID	Department	Years of Service	Performance Score
1	Sales	5	85
2	HR	3	92
3	Marketing	7	78
4	Sales	4	90
5	HR	2	76

1. Create a line chart to visualize the performance trend of employees over time. Include a legend and labels.
2. Generate a bar chart showing the distribution of employees across different departments. Label the chart elements.
3. Build a scatter plot to analyze the correlation between years of service and performance scores. Explain any insights.
4. Develop a Tableau dashboard with the line chart and bar chart for interactive exploration.

Set 4: Product Inventory Management

Dataset: Product Inventory

Product ID	Product Name	Quantity Available
1	Product A	250
2	Product B	175
3	Product C	300
4	Product D	200
5	Product E	220

1. Create a bar chart to visualize the quantity of each product in the inventory. Label the axes and title the chart.
2. Generate a stacked bar chart to show the quantity of each product within different product categories.
3. Build a scatter plot to explore the relationship between product price and quantity available. Explain the findings.
4. Develop a Tableau dashboard with the bar chart and stacked bar chart to allow users to interact with the data.

Set 5: Website Analytics

Dataset: Website Traffic

Date	Page Views	Click-through Rate
2023-01-01	1500	2.3%
2023-01-02	1600	2.7%
2023-01-03	1400	2.0%

2023-01-04	1650	2.4%
2023-01-05	1800	2.6%

1. Create a line chart to visualize the trend in daily page views over time. Label the axes and title the chart.
2. Generate a bar chart showing the top N days with the highest click-through rates. Label the chart elements.
3. Develop a stacked area chart to display the distribution of user interactions (likes, shares, comments) on a website.
4. Build a Tableau dashboard with the line chart and bar chart for interactive exploration of website traffic data.

Set 6

Product Sales Analysis

Dataset: Monthly Product Sales

Product ID	Product Name	January Sales	February Sales	March Sales
1	Product A	2000	2200	2400
2	Product B	1500	1800	1600
3	Product C	1200	1400	1100

1. Create a grouped bar chart to visualize the sales of each product for the first quarter. Label the chart elements.
2. Generate a stacked area chart to represent the overall sales trend for all products over the three months.
3. Build a table to show the monthly sales data for each product. Label the table elements.
4. Develop a Tableau dashboard combining the grouped bar chart, stacked area chart, and the table for interactive exploration of sales data.

Set 7 Customer Demographics Analysis

Dataset: Customer Demographics

Customer ID	Age	Gender	Income (in \$)
1	28	Female	50000
2	35	Male	60000
3	42	Female	75000

1. Create a bar chart to visualize the distribution of customer ages. Label the axes and title the chart.
2. Generate a pie chart to represent the distribution of customers by gender.
3. Build a table to show the demographic information for each customer. Label the table elements.
4. Develop a Tableau dashboard combining the bar chart, pie chart, and the table for interactive exploration of customer demographics.

Set 8 Employee Performance Analysis

Dataset: Employee Performance

Employee ID	Department	Years of Service	Performance Score
1	Sales	5	85

2	HR	3	92
3	Marketing	7	78

1. Create a line chart to visualize the performance trend of employees over time. Label the axes and title the chart.
2. Generate a bar chart showing the distribution of employees across different departments. Label the chart elements.
3. Build a table to display the performance data for each employee. Label the table elements.
4. Develop a Tableau dashboard combining the line chart, bar chart, and the table for interactive exploration of employee performance data.

set 9

Dataset: Online_Learning_Activity.csv

Student_ID	Gender	Age	Course	Study_Time (hrs)	Videos_Watched	Quiz_Score	Login_Date
L01	Male	20	R	3.5	12	78	2025-01-05
L02	Female	22	R	4.2	15	85	2025-01-05
L03	Male	19	SQL	2.0	8	65	2025-02-08
L04	Female	21	R	5.0	18	92	2025-02-08
L05	Male	23	R	2.5	9	70	2025-03-12
L06	Female	20	SQL	4.0	14	88	2025-03-12

1. Import the dataset in **R**. Plot a histogram for Quiz_Score. Draw a boxplot of Quiz_Score by Course. Add title and labels. Interpret student performance.
2. Create a scatter plot **using R programming** between Study_Time and Quiz_Score. Represent Videos_Watched using bubble size. Apply transparency. Explain the learning pattern.
3. Using **R programming**, convert Login_Date into Date format. Calculate average quiz score per month. Plot a line chart. Apply moving average smoothing. Identify the trend.
4. Import Online_Learning_Activity.csv **into Tableau**. Create a bar chart showing average quiz score by Course. Create a scatter plot (Study Time vs Quiz Score). Add filter for Gender and Course. Design a mini interactive dashboard.

Set 10 Survey Responses Analysis

Dataset: Survey Responses

Survey ID	Question 1	Question 2	Question 3
1	A	B	C
2	B	A	D
3	C	A	B

1. Create a grouped bar chart to visualize the distribution of answers for Question 1. Label the chart

elements.

2. Generate a stacked bar chart to represent the overall distribution of responses for all three questions.
3. Build a table to show the survey response data for each survey. Label the table elements.
4. Develop a Tableau dashboard combining the grouped bar chart, stacked bar chart, and the table for interactive exploration of survey responses.

Set 11

Product Category Analysis

Dataset: Product Categories

Category	Sales (in \$)
Electronics	50000
Clothing	35000
Appliances	40000

1. Generate a funnel chart to analyze the sales conversion process for each product category. Label the stages and title the chart.
2. Build a table to display the sales data for each product category. Label the table elements.
3. Develop a Tableau dashboard combining the pie chart, funnel chart, and the table for interactive exploration of product category data.
4. Create a pie chart to represent the distribution of sales across product categories. Include labels.

Set 12 :Dataset: Website Traffic

Date	Page Views	Click-through Rate
2023-01-01	1500	2.3%
2023-01-02	1600	2.7%
2023-01-03	1400	2.0%

1. Create a line chart to visualize the trend in daily page views over time. Label the axes and title the chart.
2. Generate a bar chart showing the top N days with the highest click-through rates. Label the chart elements.
3. Build a table to show the website traffic data for each day. Label the table elements.
4. Develop a Tableau dashboard combining the line chart, bar chart, and the table for interactive exploration of website traffic data.

13. Dataset: Geographic Data

City	Population	Avg. Temperature	Elevation
City A	500000	75	1000
City B	700000	68	800
City C	600000	80	1200

1. Create a map chart to visualize the distribution of cities on a geographic map. Label the map

elements.

2. Generate a scatter plot to explore the relationship between average temperature and population size. Explain any insights.
3. Build a table to display the geographic data for each city. Label the table elements.
4. Develop a Tableau dashboard combining the map chart, scatter plot, and the table for interactive exploration of geographic data.

14. Dataset: Survey Results

Respondent	Question 1	Question 2	Question 3
1	A	B	C
2	B	A	D
3	C	A	B

1. Create a stacked bar chart to display the distribution of answers for Question 1. Label the chart elements.
2. Generate a radar chart to visualize the overall pattern of responses across all three questions.
3. Build a table to show the survey response data for each respondent. Label the table elements.
4. Develop a Tableau dashboard combining the stacked bar chart, radar chart, and the table for interactive exploration of survey responses.

Set 15: Product Sales Analysis

Dataset: Monthly Product Sales

Product ID	Product Name	January Sales	February Sales	March Sales
1	Product A	2000	2200	2400
2	Product B	1500	1800	1600
3	Product C	1200	1400	1100

1. Create a grouped bar chart to visualize the sales of each product for the first quarter. Label the chart elements.
2. Generate a stacked area chart to represent the overall sales trend for all products over the three months.
3. Build a table to show the monthly sales data for each product. Label the table elements.
4. Develop a Tableau dashboard combining the grouped bar chart, stacked area chart, and the table for interactive exploration of sales data.

Set 16: Customer Demographics Analysis

Dataset: Customer Demographics

Customer ID	Age	Gender	Income (in \$)
1	28	Female	50000
2	35	Male	60000
3	42	Female	75000

1. Create a bar chart to visualize the distribution of customer ages. Label the axes and title the chart.
2. Generate a pie chart to represent the distribution of customers by gender.
3. Build a table to show the demographic information for each customer. Label the table elements.
4. Develop a Tableau dashboard combining the bar chart, pie chart, and the table for interactive exploration of customer demographics.

Set 17: Employee Performance Analysis
Dataset: Employee Performance

Employee ID	Department	Years of Service	Performance Score
1	Sales	5	85
2	HR	3	92
3	Marketing	7	78

1. Create a line chart to visualize the performance trend of employees over time. Label the axes and title the chart.
2. Generate a bar chart showing the distribution of employees across different departments. Label the chart elements.
3. Build a table to display the performance data for each employee. Label the table elements.
4. Develop a Tableau dashboard combining the line chart, bar chart, and the table for interactive exploration of employee performance data.

Set 18: Product Inventory Management
Dataset: Product Inventory

Product ID	Product Name	Quantity Available	Price (in \$)
1	Product A	250	20
2	Product B	175	15
3	Product C	300	18

1. Create a bar chart to visualize the quantity of each product in the inventory. Label the axes and title the chart.
2. Generate a stacked bar chart to show the quantity of each product within different product categories.
3. Build a table to show the inventory data for each product. Label the table elements.
4. Develop a Tableau dashboard combining the bar chart, stacked bar chart, and the table for interactive exploration of inventory data.

Set 19: Survey Responses Analysis
Dataset: Survey Responses

Survey ID	Question 1	Question 2	Question 3
1	A	B	C
2	B	A	D
3	C	A	B

1. Create a grouped bar chart to visualize the distribution of answers for Question 1. Label the chart elements.
2. Generate a stacked bar chart to represent the overall distribution of responses for all three questions.
3. Build a table to show the survey response data for each survey. Label the table elements.
4. Develop a Tableau dashboard combining the grouped bar chart, stacked bar chart, and the table for interactive exploration of survey responses.

Set 20: Stock Analysis
Dataset: Stock Prices

Date	Stock A	Stock B	Stock C
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2023-01-01	100	150	120
2023-01-02	105	152	118
2023-01-03	110	148	122

1. Create a line chart to visualize the stock prices of three companies (Stock A, Stock B, and Stock C) over a specific time period. Label the axes and title the chart.
2. Generate a bar chart showing the daily percentage change in stock prices for Stock A. Label the chart elements.
3. Build a table to display the stock price data for each company over the given period. Label the table elements.
4. Develop a Tableau dashboard combining the line chart, bar chart, and the table for interactive exploration of stock prices.

Set 21

Dataset: Energy_Consumption_Data.csv

Sector	Region	Month	Temperature (°C)	Units_Consumed (kWh)	Cost (₹)	Renewable_Usage (%)	Peak_Hours
Residential	North	Jan	15	320	2100	22	4
Commercial	South	Jan	24	540	3600	18	6
Industrial	West	Feb	20	880	5900	12	8
Residential	East	Feb	18	350	2300	25	5
Commercial	North	Mar	28	610	4100	20	7
Industrial	South	Mar	30	920	6200	15	9

1. Import the dataset **in R**. Plot a histogram and density plot for Units_Consumed. Compare consumption patterns.
2. Using **R programming**, create a scatter plot between Temperature and Units_Consumed. Represent Peak_Hours using bubble size. Apply transparency.
3. Using **R**, calculate average Renewable_Usage for each Sector. Create a bar chart. Interpret renewable energy adoption.
4. Import Energy_Consumption_Data.csv into **Tableau**. Create a line chart for monthly energy usage trends. Create a heatmap (Sector vs Region vs Units). Add filters for Month and Sector. Design an interactive dashboard.

Set 22: Time Series Analysis

Dataset: Monthly Sales Data

Month	Sales (in \$)
January	15000
February	18000
March	22000
April	20000

May	23000
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1. In R, create a time series line chart to visualize the trend in monthly sales. Label the axes and title the chart.
2. In R, generate a scatter plot to analyze the relationship between advertising budget and monthly sales. Explain any insights.
3. In Tableau, build a dashboard combining a line chart showing sales trend and a pie chart displaying the distribution of sales across products.
4. In R, create an autocorrelation plot to identify seasonality in the time series data.

Set 23: Employee Performance Analysis

Dataset: Employee Performance

Employee ID	Department	Years of Service	Performance Score
1	Sales	5	85
2	HR	3	92
3	Marketing	7	78
4	Sales	4	90
5	HR	2	76

1. In R, create a bar chart to visualize the distribution of employees across different departments. Label the chart elements.
2. In R, generate a line chart to visualize the performance trend of employees over time. Label the axes and title the chart.
3. In Tableau, build a dashboard combining a bar chart showing department distribution and a scatter plot displaying the relationship between years of service and performance scores.
4. In R, develop a table showing the performance data for each employee. Label the table elements.

Set 24 Product Inventory Management

Dataset: Product Inventory

Product ID	Product Name	Quantity Available
1	Product A	250
2	Product B	175
3	Product C	300
4	Product D	200
5	Product E	220

1. In R, create a bar chart to visualize the quantity of each product in the inventory. Label the axes and title the chart.
2. In R, generate a stacked bar chart to show the quantity of each product within different product categories.
3. In Tableau, build a dashboard combining the bar chart and stacked bar chart for interactive exploration of inventory data.
4. In R, create a scatter plot to explore the relationship between product price and quantity available. Explain any insights.

Set 25: Website Traffic Analysis

Dataset: Website Traffic

Date	Page Views	Click-through Rate
2023-01-01	1500	2.3%

2023-01-02	1600	2.7%
2023-01-03	1400	2.0%
2023-01-04	1650	2.4%
2023-01-05	1800	2.6%

1. In Tableau, create a line chart to visualize the trend in page views over time. Label the axes and title the chart.
2. In R, generate a bar chart to show the top N days with the highest click-through rates. Label the chart elements.
3. In R, build a stacked area chart to display the distribution of user interactions (likes, shares, comments) on the website.
4. In Tableau, create a dashboard with an interactive map showing traffic sources and a bar chart displaying page views by source.

SET 26

Dataset: Student_Mini_Data.csv

Student_ID	Gender	Study_Hours	Attendance	Math_Score	Science_Score	Exam_Date
S01	Male	2.0	78	62	65	2025-01-10
S02	Female	3.5	90	80	85	2025-01-10
S03	Male	1.5	70	55	58	2025-02-12
S04	Female	4.0	95	90	92	2025-02-12
S05	Male	2.8	85	72	74	2025-03-15
S06	Female	3.0	92	82	86	2025-03-15

1. Import the dataset **in R**. Plot a histogram for Math_Score. Draw a boxplot of Science_Score by Gender. Add proper title and labels. Interpret the results.
2. Create a scatter plot **using R programming** between Study_Hours and Math_Score. Use different colors for Gender. Add a regression line. Explain the relationship between study time and performance.
3. Using **R**, Convert Exam_Date into Date format. Calculate monthly average Math scores. Plot a line chart for the trend. Apply moving average smoothing. Identify any increasing or decreasing pattern.
4. Import Student_Mini_Data.csv into **Tableau**. Create a bar chart showing average Math scores by Gender. Create a line chart for score trends over time. Add a filter for Exam_Date. Design a simple interactive dashboard.

SET 27.

Patient Health Risk Analysis

Patient_ID	Age	BMI	BP	Cholesterol
P1	25	22	120	180
P2	40	28	135	210
P3	55	30	145	240
P4	35	26	130	200
P5	60	32	150	260

1. Using R, create a scatterplot matrix for Age, BMI, BP, and Cholesterol. Examine pairwise relationships to identify strong correlations, clusters, or potential risk patterns.
2. Construct Q-Q plot and ECDF for Cholesterol levels to assess distribution behaviour. Comment on normality assumption and skewness.
3. Using R, create a bar chart showing the average values of each health indicator (Age, BMI, BP, Cholesterol). Include axis labels, a descriptive title, and distinct colors for each bar. How do these averages relate to general patient health risk
4. Create an interactive Tableau dashboard with a heatmap highlighting high-risk patients and an Age filter for exploring risk patterns. Include hover tooltips showing all health indicators. Analyze which patients are highest risk, how age affects risk, and how this dashboard can aid healthcare professionals in prioritizing care.

SET 28. Vehicle Performance Analysis

Dataset:

Vehicle_ID	Engine_Size (L)	Horsepower	Fuel_Efficiency (km/l)	Top_Speed (km/h)	Safety_Rating
V1	1.5	110	18	180	4
V2	2.0	150	15	200	5
V3	3.0	250	12	250	5
V4	2.5	200	14	220	4
V5	1.8	130	17	190	3

1. Using R, develop a violin plot to compare the distribution of Fuel_Efficiency across different Safety_Rating categories. Examine the spread, central tendency, and variability within each group, and interpret what this indicates about efficiency patterns.
2. Using R, construct a scatter plot to explore the relationship between Horsepower and Top_Speed, applying color differentiation based on Engine_Size. Analyze the strength and direction of correlation and find observable performance trends.
3. Using R, create a correlation heatmap for all numerical variables in the dataset. Identify the variables most strongly associated with Top_Speed and their influence on overall vehicle performance.
4. In Tableau, design an interactive dashboard incorporating bar charts and scatter plots, along with filters for Engine_Size and Safety_Rating, to highlight vehicles that balance fuel efficiency and high performance. Provide analytical interpretation of the insights derived from the dashboard.

SET 29 .Student Academic Performance

Dataset:

Student_ID	Age	Study_Hours	Attendance (%)	Test_Score	Participation_Score
S1	19	12	90	85	8
S2	21	8	70	70	7
S3	20	15	95	92	9
S4	22	10	85	80	8
S5	23	7	60	65	6

1. Using R, create a stacked area chart showing Test_Score and Participation_Score across students. Analyze trends.
2. Using R, generate a boxplot of Study_Hours grouped by Attendance quartiles. Interpret patterns. Include axis labels, title, and colors for each quartile. Analyze the boxplot to identify median and interquartile range of study hours per attendance quartile and any outliers in study hours.
3. Using R, construct a density plot for Test_Score to evaluate score distribution and outliers.

4. In Tableau, design an interactive scatterplot with Study_Hours on X-axis, Test_Score on Y-axis, colored by Participation_Score. Include filters for Age.

SET 30

Dataset: Mobile_App_Usage.csv

User_ID	Gender	Age	Screen_Time (hrs)	App_Usage_Count	Data_Used (GB)	Satisfaction	Usage_Date
U01	Male	20	4.5	18	2.4	3	2025-01-08
U02	Female	22	6.0	25	3.8	5	2025-01-08
U03	Male	19	3.2	12	1.6	3	2025-02-11
U04	Female	21	7.1	30	4.5	5	2025-02-11
U05	Male	23	2.8	10	1.2	2	2025-03-14
U06	Female	20	5.4	22	3.1	4	2025-03-14

1. Import the dataset **in R**. Plot a histogram and density plot for Screen_Time. Add suitable colors and labels. Interpret the screen-time distribution.
2. Create a scatter plot using **R programming** between Data_Used and Screen_Time. Calculate the correlation between them. Add a trend line. Explain the relationship.
3. **Using R**, find the average Satisfaction score for each Gender. Create a bar chart for the averages. Add value labels. Interpret user satisfaction patterns.
4. Import Mobile_App_Usage.csv **into Tableau**. Create a bubble chart (Screen Time vs Data Used). Use color for Gender and size for App Usage Count. Create a line chart for usage trends. Add filters for Gender and Usage_Date. Design a mini dashboard.